UserArena: Benchmarking Interactive User Simulation for Long-Horizon Shopping Agents

From: Hugging Face

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The Brief

What shipped
The RecGPT-Team paper introduces UserArena, an interactive benchmark for long-horizon shopping agents, and On-Policy Trajectory Distillation (OPTD) for training user simulators.

Capabilities
UserArena contains 64,729 shopping-session trajectories from 11,010 users, covering 566,102 items and 13 action types. Its five-level evaluation covers output validity, local decisions, sequential behavior, personalization, and outcomes; the authors report that OPTD has the strongest overall fidelity among evaluated simulators and most closely reproduces a production policy effect in offline simulation.

Availability
The article links code and data at https://github.com/RecGPT-Team/UserArena and a project page at https://recgpt-team.github.io/UserArena-Page. API, deployment details, licensing, pricing, and access requirements are not stated.

Why it matters
Teams evaluating shopping agents can assess multi-step behavior and personalization against real-world trajectories, rather than relying only on next-action accuracy. The benchmark also provides offline validation against outcomes from a controlled production A/B test, and OPTD is a simulator-training approach to consider for producing more faithful rollouts.

Caveats
The article gives no numerical fidelity scores, metric definitions, or details of the A/B test, so the strength and transferability of the reported results cannot be assessed from this description. Access and usage terms for the linked code and data are not stated.

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Full Announcement Details

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